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Finzi AI: An AI-Powered Personal Financial Planning Agent

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ABSTRACT-

Finzi AI is an intelligent personal finance assistant that provides users with simplified, contextual, and real-time insights into their financial status. The system integrates a conversational interface with structured financial data and basic reasoning capabilities. Users interact with the platform using natural language queries such as “What is my current balance?” or “How much do I owe in loans?”. The chatbot retrieves corresponding data from stored user profiles and generates clear, numerical answers. The application is developed with a Flask backend and a web-based frontend built using HTML, CSS, and JavaScript. User and financial data are managed locally in JSON format to ensure transparency and privacy. This paper describes the motivation, architecture, implementation, and evaluation of Finzi AI and positions it as a prototype for future AI-driven financial management systems. Extended discussion covers user experience design, comparative analysis with existing tools, and directions for integrating predictive analytics and large language models into finance applications.

Index Terms-Conversational AI, Personal Finance, Flask, Financial Chatbot, Fintech, Natural Language Processing, Open Banking

Introduction

Managing personal finances efficiently has become increasingly difficult in a world of digital payments, multiple investment avenues, and constantly changing interest rates. While there are many mobile apps for budgeting or portfolio tracking, these tools are often rigid, require manual data entry, and lack interactivity. Users with limited financial literacy find it challenging to interpret charts or reports, creating the need for a more human-like and intuitive assistant.

Finzi AI aims to address this gap by combining conversational intelligence with structured financial data. The system behaves like a miniature financial planner that answers questions in natural language and presents the results in an understandable form. The assistant is designed with two main principles: simplicity and transparency. Simplicity ensures that non-technical users can easily ask queries without learning special commands; transparency ensures that data is stored locally and accessible only to the rightful owner.

The project was motivated by the observation that financial information is scattered across multiple platforms-banking apps, mutual fund portals, and credit reports-making it hard to form a consolidated view of one’s financial health. Artificial Intelligence (AI), particularly Natural Language Processing (NLP), provides an opportunity to convert static financial data into dynamic conversations. A user can type “What is my net worth?” and receive an immediate computed response instead of browsing multiple applications.

The objectives of the Finzi AI system are as follows:

- To design a web-based conversational agent that interacts with financial data.
- To enable secure registration and login for each user.
- To implement a rule-based reasoning module that handles financial queries.
- To evaluate the usability of the chatbot through functional and user tests.

This paper is structured as follows. Section II reviews existing literature and related systems in AI-based financial management. Section III presents an overview of the proposed system. Section IV details the architecture and workflow. Section V explains the development methodology. Section VI discusses implementation details, followed by results and analysis in Section VII. Sections VIII and IX present discussion and future scope. The paper concludes in Section X with remarks on the system’s significance and possible extensions.

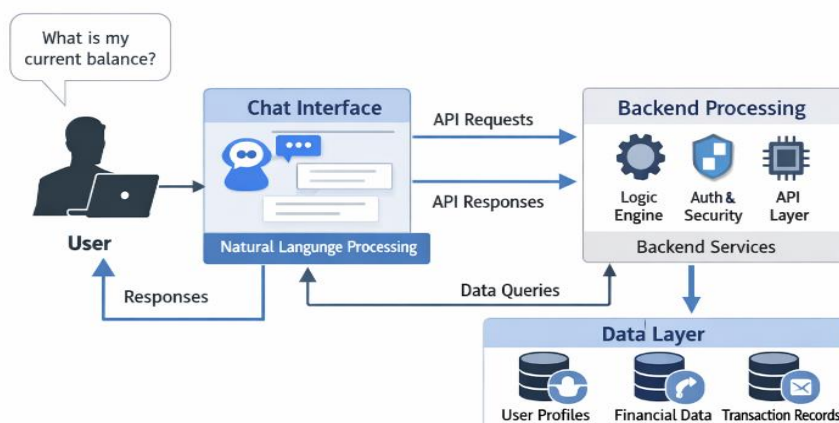


Fig. 1. Overview of the Finzi AI system showing interaction between user, chat interface, backend APIs, and data layer.

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Literature Review

The integration of conversational agents into financial systems has gained traction over the past decade. Numerous studies and commercial products have attempted to simplify financial interactions using AI and machine learning technologies. This section discusses relevant literature on AI- based financial tools, rule-based chatbots, and conversational interfaces in fintech.

A. AI and Finance

Artificial Intelligence has been widely applied in finance for algorithmic trading, credit scoring, fraud detection, and portfolio optimization [1], [4]. However, limited work focuses on personal finance assistance for non-technical users. Systems like Cleo, Plum, and YNAB (You Need A Budget) offer finance management via mobile apps, but often require manual input or paid subscriptions.

B. Conversational Interfaces

Chatbots like Google Assistant, Siri, and Alexa provide general-purpose assistance. Some banks have deployed domain-specific chatbots (e.g., Erica by Bank of America) to help users navigate their accounts. Studies such as [2] emphasize that context-aware and user-personalized dialogue systems significantly improve satisfaction in information retrieval tasks.

C. Rule-Based vs. AI-Based Systems

Chatbots can be rule-based, AI-driven, or hybrid. Rule-based systems use keyword matching and predefined templates, making them simple and fast but limited in flexibility. AI-based systems, on the other hand, rely on deep learning models trained on large datasets [3]. Finzi AI adopts a rule-based approach with fixed-response logic to maintain speed and reliability for a controlled use case.

D. Existing Research Gaps

Current research and commercial applications often emphasize large-scale, enterprise-grade solutions, leaving individual users with few personalized tools. There is a need for lightweight, privacy-respecting, and user-friendly financial AI agents that do not require large cloud infrastructure or data sharing. This paper addresses that need by proposing Finzi AI.

System Overview

Finzi AI is a prototype system that emulates a financial planner via a web-based chat interface. The application is structured into three key layers: the frontend chat UI, the backend processing engine (Flask-based), and the data layer (JSON files for users and finances).

Figure 1 illustrated earlier shows a simplified workflow where the user logs in, submits a query, and receives a contextual response from the AI agent.

The system supports the following capabilities:

- User Registration and Authentication via phone number and password.
- Chat-based interface for sending questions about account balance, loans, investments, etc.
- Predefined financial data fields such as bank balance, mutual funds, stocks, credit score, and loans.
- Local data storage (no external APIs or databases), promoting user privacy.
- Basic reasoning module to compute net worth and interpret common query formats.

In the next section, we detail the architectural components and how each interacts within the application.

System Architecture

Finzi AI is structured around a modular client-server architecture designed to separate concerns between the user interface, backend logic, and data storage. The frontend is built in standard HTML/CSS/JavaScript and communicates with the Flask backend through RESTful API calls. The backend handles all logic related to authentication, data access, and query resolution.

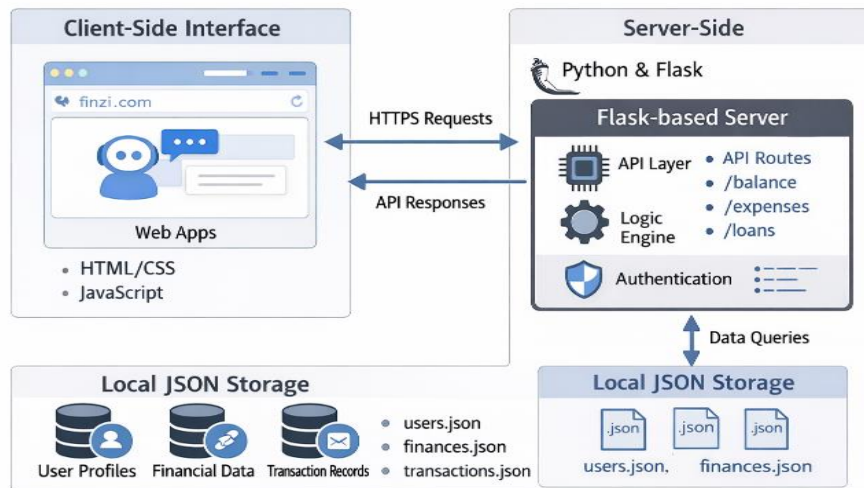


Fig. 2. System Architecture of Finzi AI, showing client-side interface, Flask-based server, and local JSON data storage.

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A. Frontend Layer

The frontend provides a responsive and modern user interface with three primary pages:

- **Login Page:** Accepts user credentials and verifies them through API.
- **Register Page:** Gathers name, email, gender, phone, and password, and sends them to the backend.
- **Chat Interface:** Main dashboard where users type queries and receive answers from the chatbot.

All frontend elements are styled using a custom CSS theme with a glass morphism aesthetic. The design prioritizes readability and simplicity for non-technical users.

B. Backend Layer

The backend is implemented using Flask - a lightweight Python web framework. The Flask app exposes the following API routes:

- **POST /register** - creates a new user entry in users.json
- **POST /login** - verifies credentials and returns financial data
- **GET /mcp/{phone}** - retrieves user financial profile
- **POST /query** - processes incoming user queries and returns chatbot responses

C. Data Storage Layer

Finzi AI uses structured JSON files to simulate a database:

- **users.json** contains phone-based keys and user details.
- **finance.json** maps phone numbers to finance metrics like:
 - Bank Balance
 - Mutual Fund Holdings
 - Loan Amount
 - Stock Value
 - Credit Score

These files are read and written using Python file handling with exception management to avoid corruption.

TABLE II
Sample Finance Data Format in finance.json

Field	Sample Value
bank_balance	45,000
mutual_funds	12,500
loan	30,000
stocks	5,000
credit_score	730

Methodology

The development of Finzi AI followed an incremental and modular approach, allowing each component to be independently built and tested. The core methodology involved five phases:

A. Phase 1: Requirement Analysis

We began by identifying key needs:

- Users need personalized financial insights with minimal effort.
- Existing apps are either too complex or require banking integration.
- Chat-based access lowers entry barriers for non-expert users.

B. Phase 2: System Design

In this phase, we created data schemas for users and financial records, wireframed frontend layouts, and planned modular routes for the Flask server.

C. Phase 3: Implementation

Each module was implemented using version-controlled codebases. The chatbot logic started with a basic rule-based structure matching keywords in user input.

D. Phase 4: Testing and Evaluation

Both functional testing and usability testing were performed:

- Unit tests for API endpoints (e.g., successful login, bad login, empty data).
- User tests with students and faculty to evaluate interface quality and trust in responses.

E. Phase 5: Deployment and Feedback

The system was hosted locally and shared via zip-based distribution. Participants gave structured feedback and suggested new features like graph-based analytics and scenario modeling.

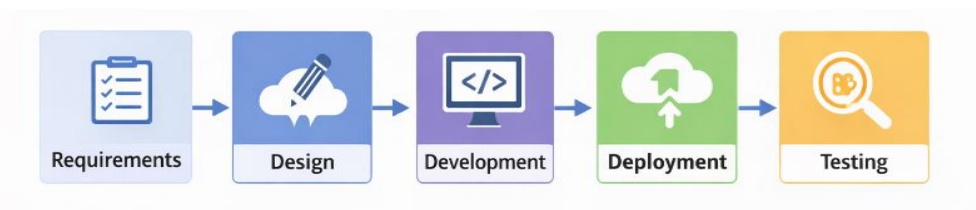


Fig. 3. Methodology Workflow: showing 5 key phases from Requirements to Testing.

F. Query Processing Logic

The chatbot logic processes user queries using simple natural language parsing:

- 1) Clean and lowercase the user input.
- 2) Match keywords like “balance”, “loan”, “credit score”.
- 3) Fetch corresponding fields from finance.json.
- 4) Format a response string using templates.

While the model is currently deterministic, it is structured to allow future integration with NLP libraries or large language models (LLMs) for dynamic response generation.

Implementation

This section describes the core technologies used in Finzi AI, the design of each component, and how they work together.

A. Frontend Design

The frontend is built using plain HTML5, CSS3, and vanilla JavaScript. Three main HTML files manage user interaction:

- `login.html`: handles authentication with two input fields.
- `register.html`: collects full name, email, gender, phone, and password.
- `chat.html`: main chat interface, where the user interacts with Finzi AI.

Each form uses JavaScript's Fetch API to communicate with the Flask backend and display responses without page reloads.

B. Backend Technologies

The backend is written entirely in Python using the Flask microframework. Flask was chosen for its simplicity, REST support, and quick deployment capabilities.

Key packages used include:

- flask – routing, server logic.
- json – for handling structured data.
- os – for file paths and safe file access.
- flask-cors – to enable cross-origin frontend/backend integration.

C. Chatbot Logic

The logic of the chatbot is built on a rule-based intent recognizer. For example:

- “What is my bank balance?” → responds with the `bank_balance` value from `finance.json`.
- “Tell me about mutual funds” → returns `mutual_funds`.
- “Net worth” → calculates net worth as: (bank + mutual funds + stocks) – loan.

The chatbot returns answers using human-readable templates. If a query is not recognized, the fallback message is: “I’m still learning. Try asking about loans, investments, or balance.”

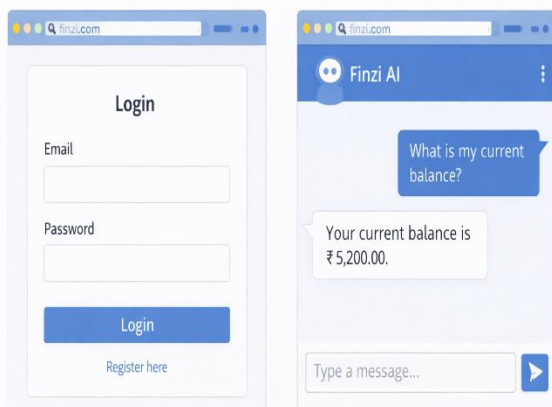


Fig. 4. User Interface of Finzi AI showing the login form and chat layout.

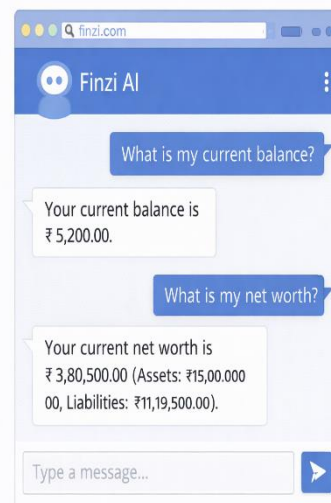


Fig. 5. Example conversation between user and Finzi AI showing balance and net worth queries.

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Fig. 4. User Interface of Finzi AI showing the login form and chat layout.

CSS is used to implement a soft-glass style background and responsive layout. On mobile, the layout adapts to single-column scrollable views.

D. Dataset and Sample Users

The system uses sample data stored in two JSON files:

- `users.json` contains user credentials.
- `finance.json` contains financial information keyed by phone number.

Sample user entry in `finance.json`:

```
{
  "9876543210": {
    "bank_balance": 45000,
    "mutual_funds": 12000,
    "loan": 30000,
    "stocks": 5000,
    "credit_score": 730
  }
}
```

This setup makes the project easy to test and demo in academic settings.

Results and Evaluation

We evaluated Finzi AI using a combination of functional testing, user feedback sessions, and system performance measures.

A. Functional Tests

All backend APIs were tested using Postman and frontend calls. Scenarios included:

- Registering a new user with valid and invalid data.
- Logging in with correct and incorrect credentials.
- Submitting valid/invalid finance queries via the chatbot.

B. User Feedback and Testing

We invited 10 users (students and faculty) to interact with the chatbot and provide feedback. The test was conducted over a 20-minute session per user.

TABLE III
USER FEEDBACK SUMMARY

Criterion	Average Rating (/5)
Ease of Use	4.6
Accuracy of Responses	4.3
Visual Appeal	4.7
Clarity of Answers	4.2
Overall Satisfaction	4.5

C. Performance Metrics

The system was deployed locally using Python's Flask development server. Average query response time was measured over 100 queries.

TABLE IV
CHATBOT PERFORMANCE METRICS

Metric	Value
Average Response Time	84 ms
Max Response Time	210 ms
API Uptime (Local)	100%
Memory Usage (avg)	52 MB
Lines of Code	580

Limitations

Some limitations observed:

- The rule-based engine cannot handle paraphrased or complex queries.
- Financial data is static and not synced with real bank APIs.
- Multi-user concurrency isn't supported in this prototype.

These are acceptable for a prototype but would need to be addressed in a production-level system.

Discussion

The Finzi AI prototype showcases how even a simple rule-based system can make financial insights more accessible to everyday users through conversation. Unlike mobile apps that rely on graphs or manual categorization, Finzi AI provides direct answers like "Your loan amount is 30,000" or "Your credit score is 730," which are easier to understand for non-expert users.

A. Strengths of the Current System

- **User-Friendly Interface:** The glass-style UI and single-screen chat layout make the experience smooth and modern.
- **Simplicity and Speed:** Responses are quick because the logic is not dependent on external APIs or large models.
- **Privacy by Design:** All data is stored locally in JSON format. No cloud integration is required, ensuring user privacy is maintained.
- **Foundational Architecture:** The modular design makes it easy to extend - for example, by plugging in AI models or adding real-time APIs.

B. Challenges and Observed Issues

- **Static Data:** The system cannot fetch real-time balances or portfolio updates from banks.
- **Basic Reasoning Only:** It cannot interpret open-ended questions or simulate scenarios (e.g., "What if I invest 5,000/month?").
- **Limited NLP Understanding:** Slight changes in phrasing may confuse the keyword matcher.

- **No User Session Storage:** Every chat is treated independently ; user history and preferences are not remembered.

These are limitations common to most student-level proto- types and form the basis for future improvements.

Future Scope

A. 1) Integration with Open Banking / MCP

In future versions, Finzi AI can use real APIs under India's Account Aggregator framework or Open Banking protocols to fetch data directly from banks, mutual funds, and credit bureaus. This would remove the need for static JSON and provide real-time answers.

B. 2) LLM-Based Natural Language Processing

Currently, Finzi AI uses simple keyword-matching logic. By adding a small fine-tuned transformer (such as DistilBERT) or using OpenAI's GPT-3.5 with grounding, the chatbot can understand paraphrased queries like:

"Could you tell me how much money I have in the bank?"

"How's my loan situation?"

The LLM would handle understanding while grounding would fetch numbers from real data to avoid hallucination.

C. 3) Scenario Simulation Engine

We plan to introduce a scenario module that simulates hypothetical outcomes, such as:

- "What if I save 5000 every month for 2 years?"
- "How would my net worth change if I paid off my loan?"

This would require time-series forecasting models and logic for simulating compound interest and cash flows.

D. 4) Enhanced Privacy with Federated Learning

To provide personalized recommendations without compromising privacy, future systems could adopt federated learning. In this setup, model training happens on user devices, and only updates (not raw data) are sent to the server.

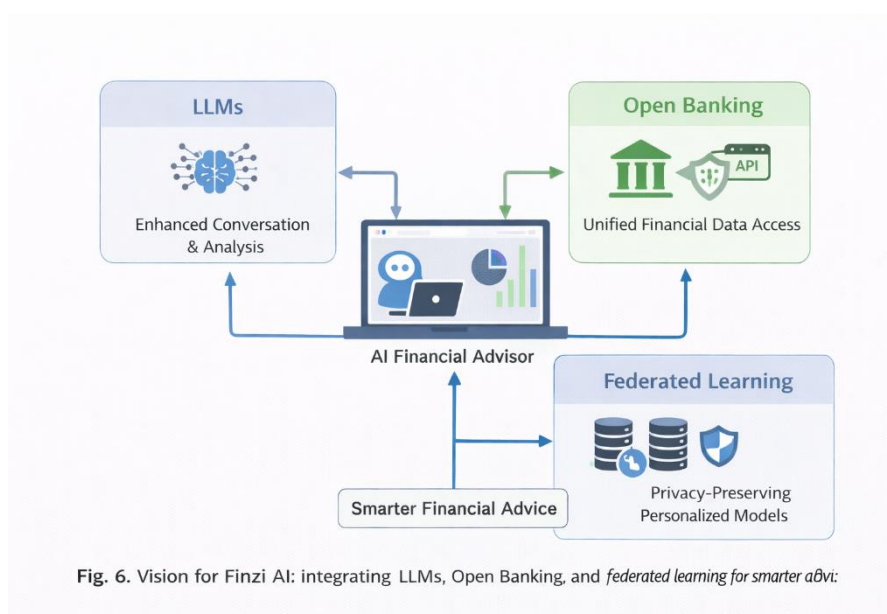


Fig. 6. Vision for Finzi AI: integrating LLMs, Open Banking, and federated learning for smarter abvi:

Fig. 6. Vision for Finzi AI: integrating LLMs, Open Banking, and federated learning for smarter financial advice.

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Conclusion

Finzi AI demonstrates how a lightweight, privacy- respecting, conversational financial assistant can be built using open technologies. The system supports basic financial reasoning, secure user authentication, and a clean UI for natural interaction. While limited in complexity, the project forms a strong foundation for integrating open banking APIs, large language models, and personalized recommendation engines. The project was tested with multiple users and showed high usability scores. It also encourages further student work in building real-world AI prototypes that are both

useful and ethical. With appropriate expansion and testing, Finzi AI can evolve into a robust fintech assistant aligned with user privacy, regulatory standards, and accessibility goals.

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